

Choosing an effective Visual

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DATA 110



Simple Text

- Best used when there are only one or two statistics or numbers to share.
- Use as little text as possible to explain and make the number/statistic prominent.



20%

of children had a
traditional stay-at-home mom
in 2012, compared to 41% in 1970

FIGURE 2.3 Stay-at-home moms simple text makeover

Tables

- Best to use when there are several units of measure.
- Design should fade into the background, data should standout.

Heavy borders

Group	Metric A	Metric B	Metric C
Group 1	\$X.X	Y%	Z,ZZZ
Group 2	\$X.X	Y%	Z,ZZZ
Group 3	\$X.X	Y%	Z,ZZZ
Group 4	\$X.X	Y%	Z,ZZZ
Group 5	\$X.X	Y%	Z,ZZZ

Light borders

Group	Metric A	Metric B	Metric C
Group 1	\$X.X	Y%	Z,ZZZ
Group 2	\$X.X	Y%	Z,ZZZ
Group 3	\$X.X	Y%	Z,ZZZ
Group 4	\$X.X	Y%	Z,ZZZ
Group 5	\$X.X	Y%	Z,ZZZ

Minimal borders

Group	Metric A	Metric B	Metric C
Group 1	\$X.X	Y%	Z,ZZZ
Group 2	\$X.X	Y%	Z,ZZZ
Group 3	\$X.X	Y%	Z,ZZZ
Group 4	\$X.X	Y%	Z,ZZZ
Group 5	\$X.X	Y%	Z,ZZZ

FIGURE 2.4 Table borders

Heatmaps

- Good for combining different types of measurements.
- Helps understand data points relative to each other.
- Should always contain a legend.

Table

	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

Heatmap

LOW-HIGH

	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

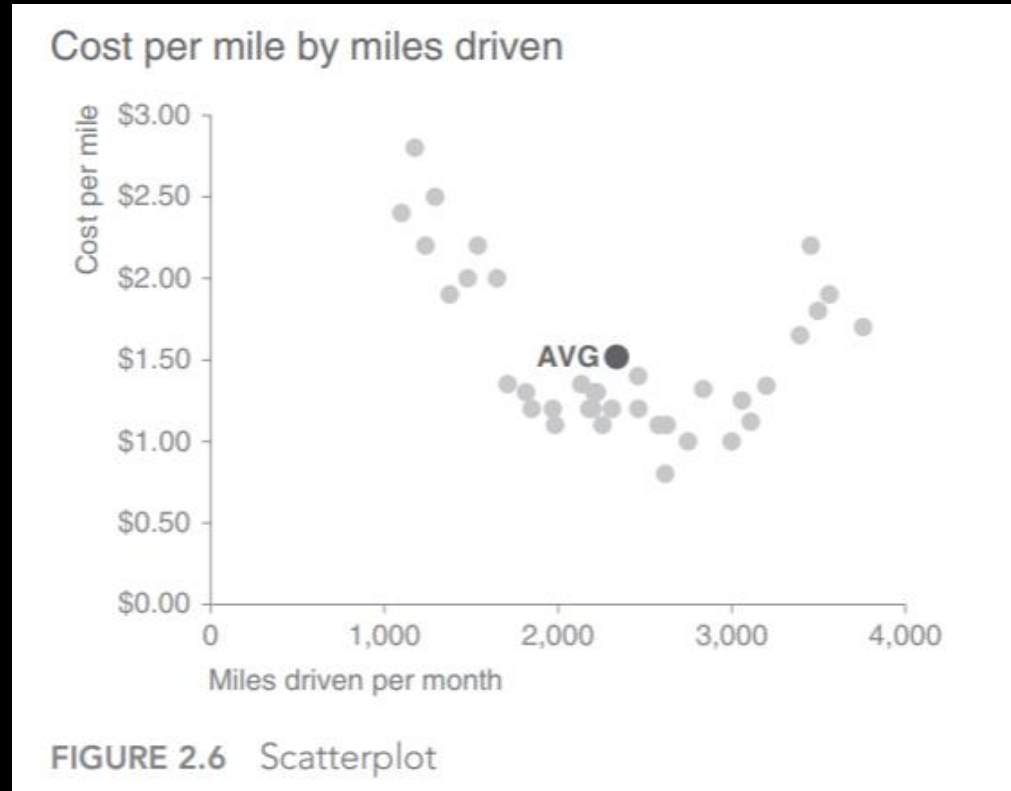
FIGURE 2.5 Two views of the same data

Graphs

- Present information to our visual system which is faster at understanding information.
- There are four types of graphs:
 - Point graphs
 - Line graphs
 - Bar graphs
 - Area graphs

Point Graphs

- Most common type is the scatterplot.
- Show data on a vertical and horizontal plane which allows you to see whether there is any relationship between the variables on each axis.

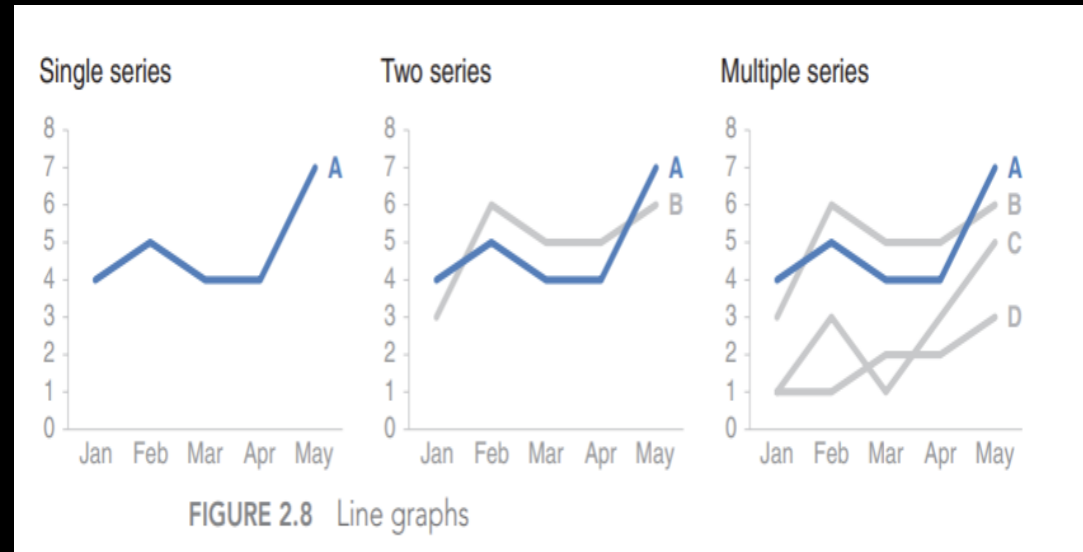


Line graphs

- Used for continuous data, implying a connection between the data.
- Types:
 - Line graphs
 - Slope graphs

Line graph

- Can plot a single or multiple series of data.
- Time intervals should be consistent.



Slope graph

- Good for showing relative increases or decreases over a period of time.
- Can plot multiple categories of data.

Employee feedback over time

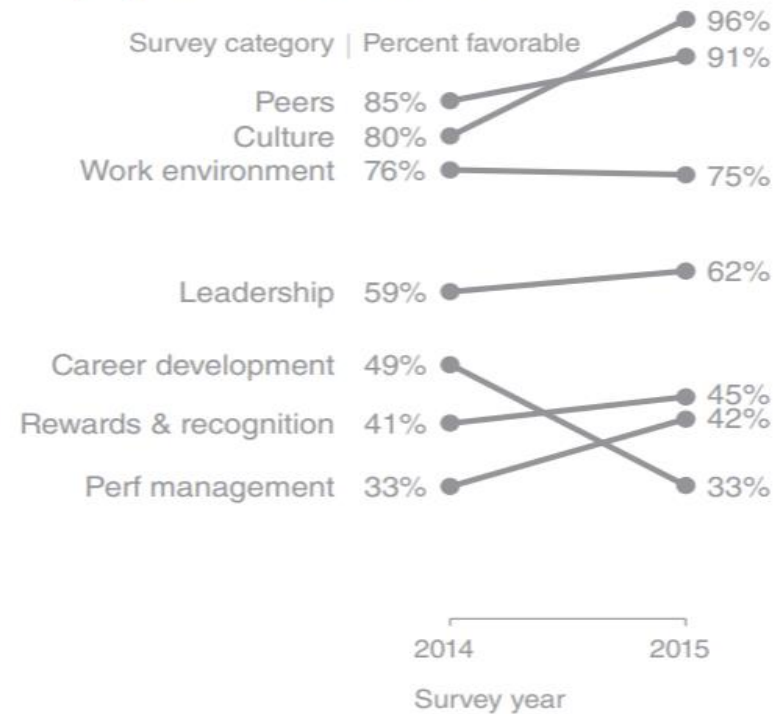


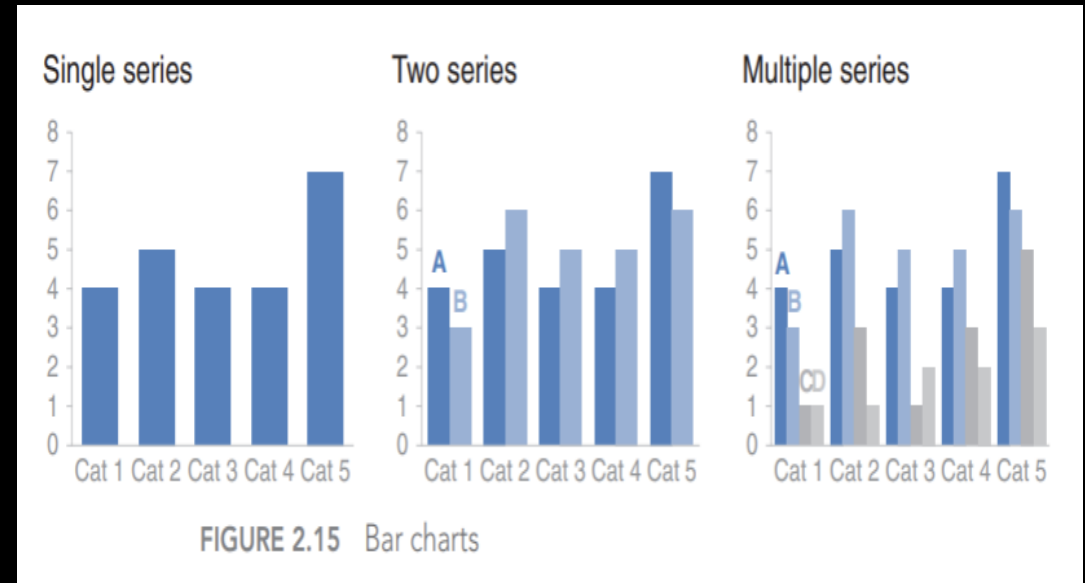
FIGURE 2.10 Slopegraph

Bar graphs

- Bar graphs are very common, which is why people understand them so easily. This is an advantage because the audience will focus more on the information presented rather than trying to understand what the graph says.
- Bars allow for easy visual comparison.
- Types:
 - Vertical bar chart
 - Stacked vertical bar chart
 - Waterfall chart
 - Horizontal bar chart
 - Stacked horizontal bar chart

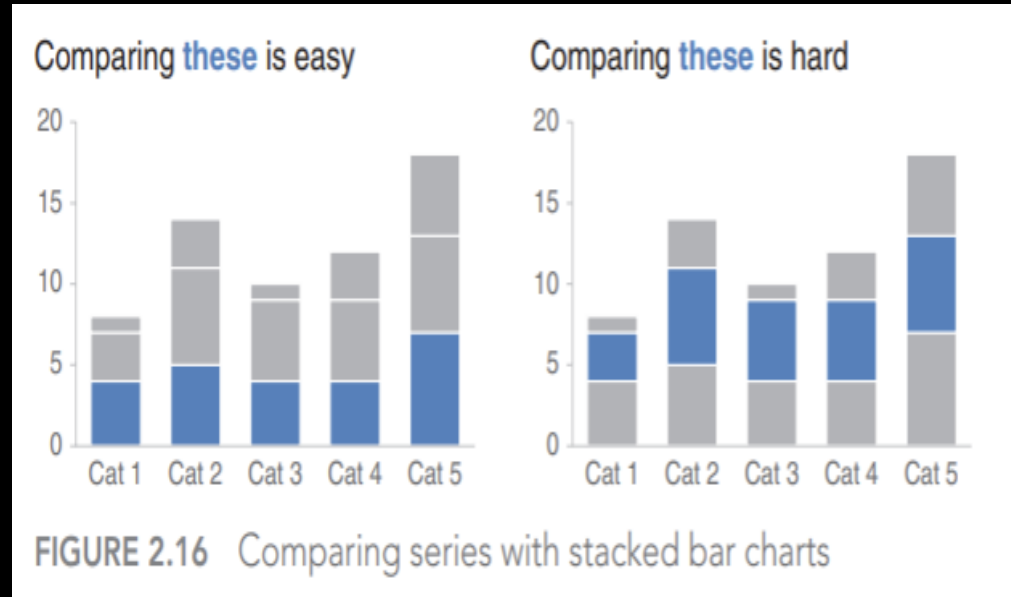
Vertical bar chart

- Can plot single or multiple series of data.
- Better not to present too many series of data because it can clutter the graph and make it difficult to understand.



Stacked vertical bar chart

- Compare categories and subcomponents within categories.
- Percentages within categories will usually add up to 100%.



Waterfall chart

- Can be used to examine pieces of a stacked bar chart to show individual increases or decreases.

2014 Headcount math

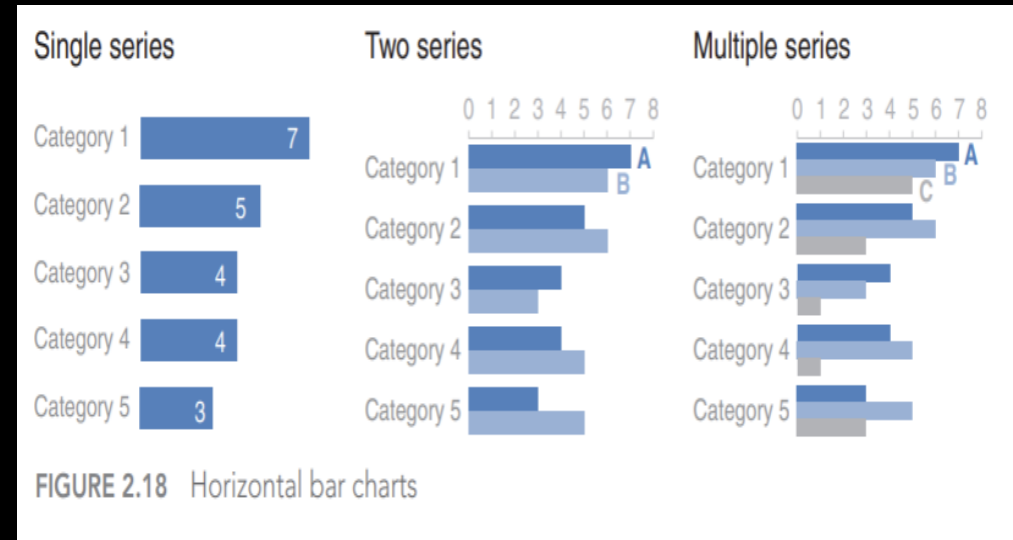
Though more employees transferred out of the team than transferred in, aggressive hiring means overall headcount (HC) increased 16% over the course of the year.



FIGURE 2.17 Waterfall chart

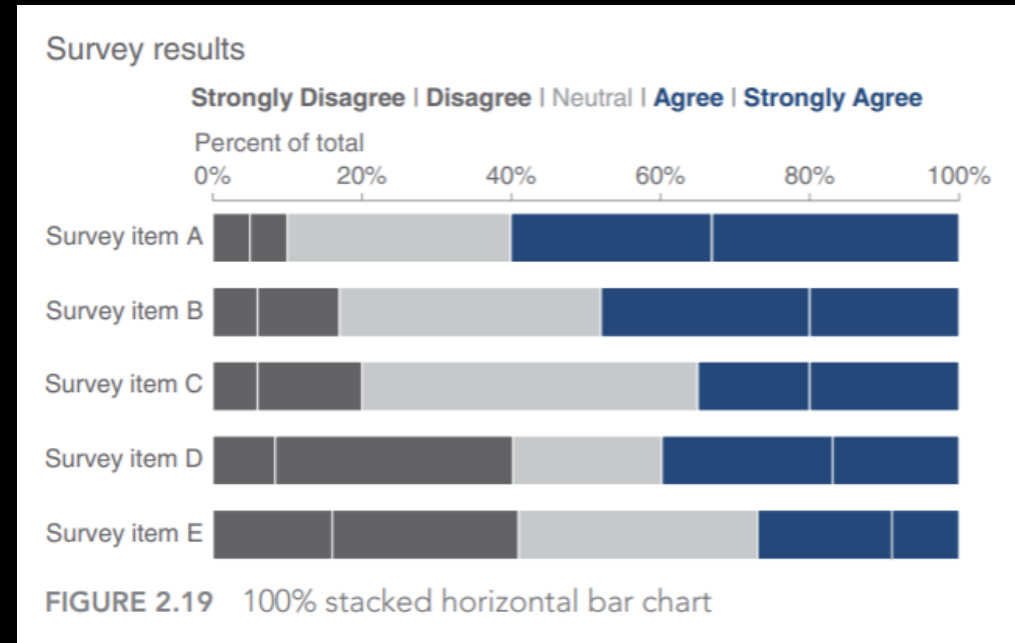
Horizontal bar chart

- Easy to read.
- Format follows natural eye movement (left to right, then down to next category).
- Convenient when category names are long.
- Can graph single or multiple series of data.



Stacked horizontal bar chart

- Shows totals across categories and percentages within categories.
- Works well with survey data.



Area chart

- Harder to read due to the need to judge the difference between different sized areas.
- Good for use when there are large differences between the values of different categories.

Interview breakdown

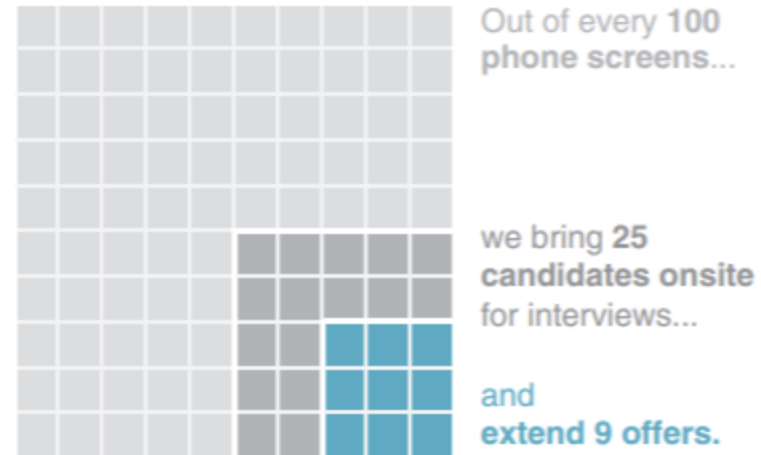


FIGURE 2.20 Square area graph

Visuals to Avoid

- Pie charts:
 - Pie charts rely on the audience being able to discern whether one area of the pie is larger than another. This can be difficult and misleading especially if the chart is not set completely flat.
- Donut charts:
 - Rely on the audience being able to tell whether one arc of the donut is longer than another. This is difficult to do and can be misleading.
- 3D:
 - Golden rule of data visualization: never use 3D. (unless plotting a third dimension)
 - Skews visual perspective of data and makes it harder to read as well as distracting.
- Second y axis:
 - Makes it difficult to understand which data points are being measured on each of the two y-axis.